RescueMedAI — Review 1 Master Study & Viva Guide

AI-Powered Disaster Response & Emergency Medical Decision Support System

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1. Executive Summary & Project Overview

RescueMedAI is an end-to-end AI decision support framework for disaster management agencies (NDMA, Red Cross). In large-scale disasters, traditional infrastructure fails due to blackouts, blocked corridors, and overwhelmed hospitals. RescueMedAI connects Computer Vision, Dynamic Graph Routing, Priority Resource Allocation, Hybrid Bayesian Triage, and Hospital Telemetry into a unified 5-phase pipeline.

> ***Key Defense Point for Viva:*** *RescueMedAI is Decision Support Software, NOT an autonomous agent. Every output carries a mandatory clinical disclaimer requiring human verification.*

2. Pipeline Architecture & Module Deep-Dive

Phase 1: Computer Vision & Severity Estimator (src/backend/vision/)

• **PyTorch MobileNetV3 (LRASPP)**: Pretrained semantic segmentation for aerial terrain.
• **Sobel Gradient Energy Filter**: G = sqrt(Gx^2 + Gy^2). If G > 22.0, terrain is flagged as structural rubble.
• **Mudflow / Water RGB Mask**: Thresholding detects submerged roads.
• **Hazard Weights**: Flooded/Blocked = 5.0 (Impassable), Severely Damaged = 3.0, Normal = 0.0.

Phase 2: Dynamic Road Network Graph & A* Routing (src/backend/routing/)

• **A* Algorithm**: f(n) = g(n) + h(n).
• **Path Cost g(n)**: Travel time + dynamic hazard penalties assessed by vision.
• **Heuristic h(n)**: Euclidean distance to goal hospital / max speed (admissible).
• **Safest vs Shortest Proof**: Demonstrates that physical shortest routes often traverse high-risk hazard zones, whereas A* finds safer detours cutting risk time significantly.

Phase 3: Emergency Resource Allocation (src/backend/allocation/)

• Matches incident location with optimal unit (ALS vs BLS ambulance) and target trauma hospital based on priority, ETA, and trauma level capabilities.

Phase 4: Hybrid Field Medical Triage Engine (src/backend/triage/)

• **Rule Engine**: Deterministic START/SALT protocol (SpO2, Respiratory Rate, AVPU).
• **Bayesian Belief Network**: Infers posterior probabilities P(Pathology | Vitals) for Tension Pneumothorax, Hemorrhagic Shock, TBI.
• **Shannon Entropy H(X)**: Quantifies diagnostic uncertainty.
• **Conservative Escalation**: Safe upgrade to RED if cumulative critical risk > 45%.

Phase 5: Hospital Pre-Arrival Telemetry (src/backend/hospital/)

• Standardized SBAR (Situation, Background, Assessment, Recommendation) clinical handoff telemetry payload and ReportLab PDF export.

3. Likely Viva Questions & Winning Answers

Q1: Why A* Search instead of Dijkstra?

Dijkstra explores uniformly in all directions f(n)=g(n). A* adds an admissible Euclidean heuristic h(n), guiding search toward the target hospital to find the optimal safest path while expanding significantly fewer nodes.

Q2: How do you handle system policies blocking PyTorch?

We built a hybrid fallback architecture. If PyTorch DLLs fail due to OS Application Control policies, the vision detector seamlessly falls back to Sobel edge gradients and color-space analysis without crashing.

Q3: What is Conservative Escalation in Triage?

To prevent fatal under-triaging, if probabilistic Bayesian inference detects cumulative critical pathology risk > 45%, the patient is safely escalated to RED even if rule thresholds weren't breached.

Q4: What is Shannon Entropy used for?

Shannon Entropy H(X) = -sum(P(x) log2 P(x)) measures uncertainty across potential pathologies. High entropy signals conflicting vitals, which penalizes the output confidence score to alert doctors.

Q5: Why is shortest route not always safest?

Physical shortest paths often cut through flooded or collapsed corridors. A* identifies safer detours that add slight distance but save significant risk cost and travel time.